Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [13CE64/12CE64]

SIXTH SEMESTER B.Tech./B.E EXAMINATION, JUNE – 2016

ESTIMATING COSTING & SPECIFICATIONS

Time: 3 Hours **Maximum Marks: 70**

Instructions:

1. Answer question No. 1(COMPULSORY) and any three of the remaining.
2. Missing data may be suitably assumed.
3. Follow the written dimension do not scale the drawing.

1. Accompanying sketch 1 shows the details of 3 bed room residential unit, compute the below mentioned items of work lay "CENTRE LINE METHOD". **28**

a. Earth work excavation for foundation in hard soil. **04**

b. Providing and laying PCC 1:4:8 for foundation using 40mm and 20mm size coarse aggregates. **03**

c. Providing and constructing size stone masonary in cm 1:6 for foundation using hammer dressed stones. **08**

d. Providing and constructing burnt brick masonary in cm 1:6 for super structure (only main walls). **08**

e Computation of correct center line length. **05**

2. a. It is proposed to construct a Road for 400mts. Calculate the volume of earth work by "Mean depth method" from the following data.

Chainage in mtrs.	500	540	580	620	660	700	740	780	820	860	900
RLia mts.	61.0	60.9	60.5	60.8	60.6	60.7	61.2	61.4	61.3	61.0	60.6
Formation level	62.0	← Down gradient 1 in 200 →									

Formation width = 10mts, side slope in filling → 2H: 1V, in cutting →1.5H:1V **14**

Draw the Longitudinal section and show typical cross sections also.

3. The sketch 3 enclosed shows the L/S and C/S of a RCC Beam. Calculate the quantity of steel required for Five such beams diameter wise and also factor 5% wastage . **14**

Wt. of bar/mt.

8mm → 0.39 kg/mt., 10mm → 0.61kg/mt., 12mm → 0.89 kg/mt. ,

16mm → 1.58 kg/mt., 20mm → 2.50 kg/mt.. 25mm → 3.80 kg/mt.

4. Write a detailed technical specification for any two items of work.

a. Earth work excavation for foundation in hard soil.

b. Providing and constructing Burnt Brick masonary

c. Providing and laying AC sheet roofing with necessary accessories. (7×2=14) **14**

5. The sketch 5 shows the details of "SLAB CULVERT". Prepare a detailed estimate for below mentioned items of work only.

a. Earthwork excavation for foundation in hard soil. **03**

b. Providing and laying M20 grade concrete for roof slab and wearing coat using 20mm and down size coarse aggregates. **04**

c. Providing and constructing Burnt Brick Masonary in CM 1:6 for super structure and parapet. **07**

6. Write short notes on: (3½ ×4=14)

a. Measurement book and its importance.

b. Reconciliation statement at site.

c. Earnest Money Deposit and Security Rete.....

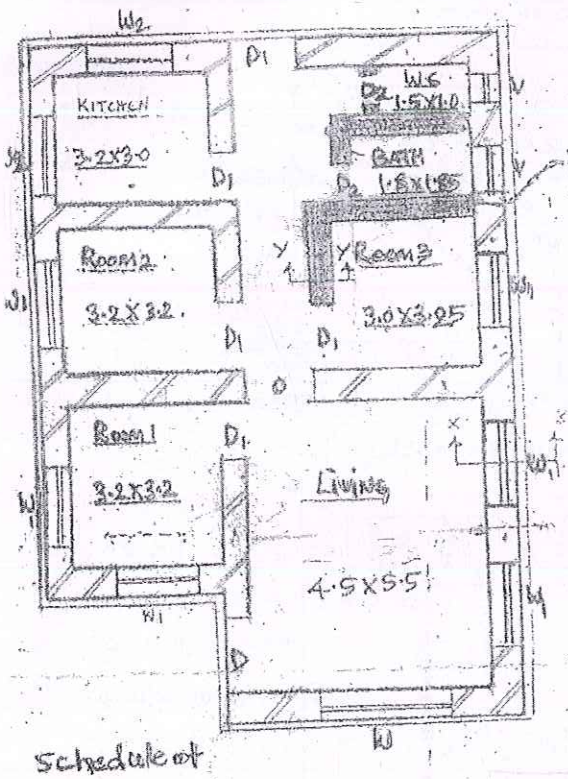
d. Mobilisation Advance. **14**

7. a. What is a contract? What are the different types of contract? **04**

b. What is Quality Assurance and Quality control? How is it evaluated in the Project site? **06**

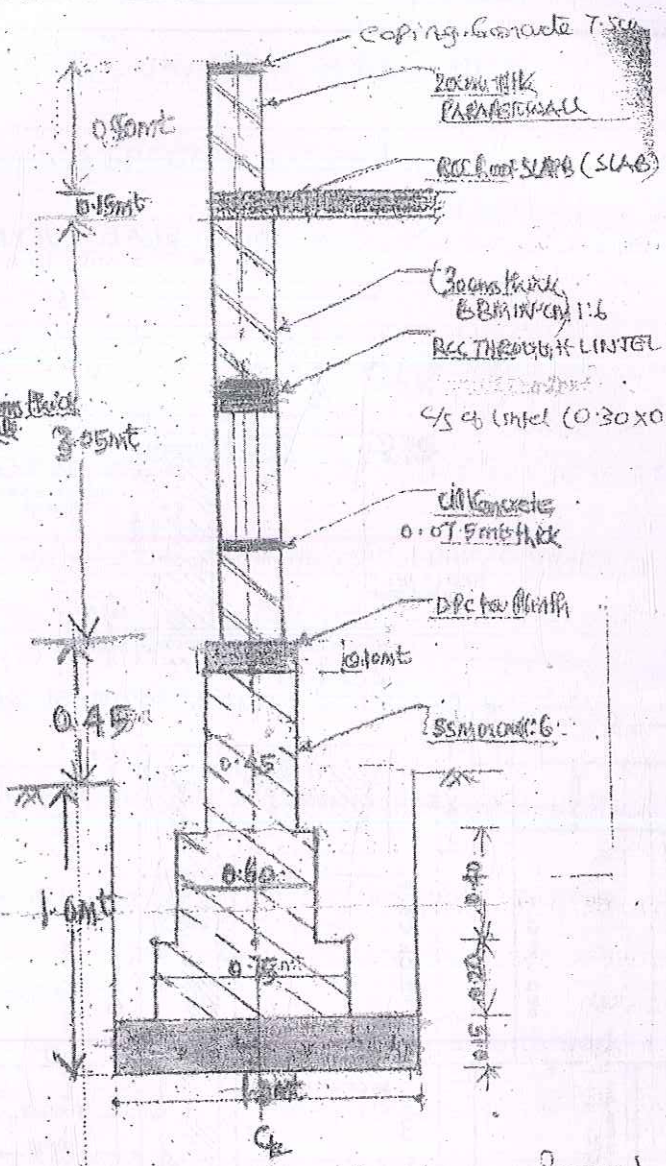
c. What is Pre bid meeting? What purpose does a pre bid meeting serve in the process of finalizing the vendor? **04**

PLAN AT SILL LEVEL

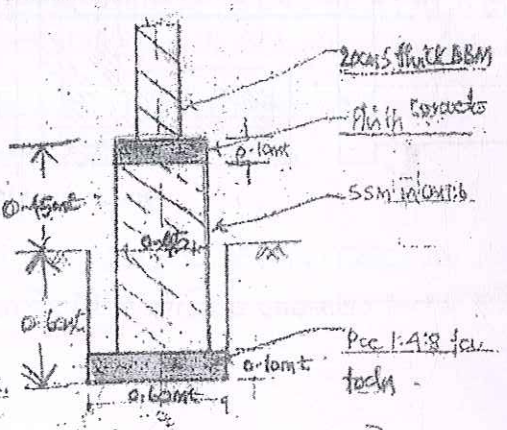


schedule of opening:-

Type	Size	Reqd No's
D	1.05 X 2.10	1
D1	0.90 X 2.10	5
Dom 'b'	0.90 X 2.10	1
D2	0.75 X 2.10	2
W	1.80 X 1.20	1
W1	1.50 X 1.20	6
W2	1.20 X 1.05	2
V	0.9 X 0.6	2



SECTION X X (MAIN WALL FOOTING)



SECTION Y-Y

PARTITION WALL FOOTING

SKETCH - 1

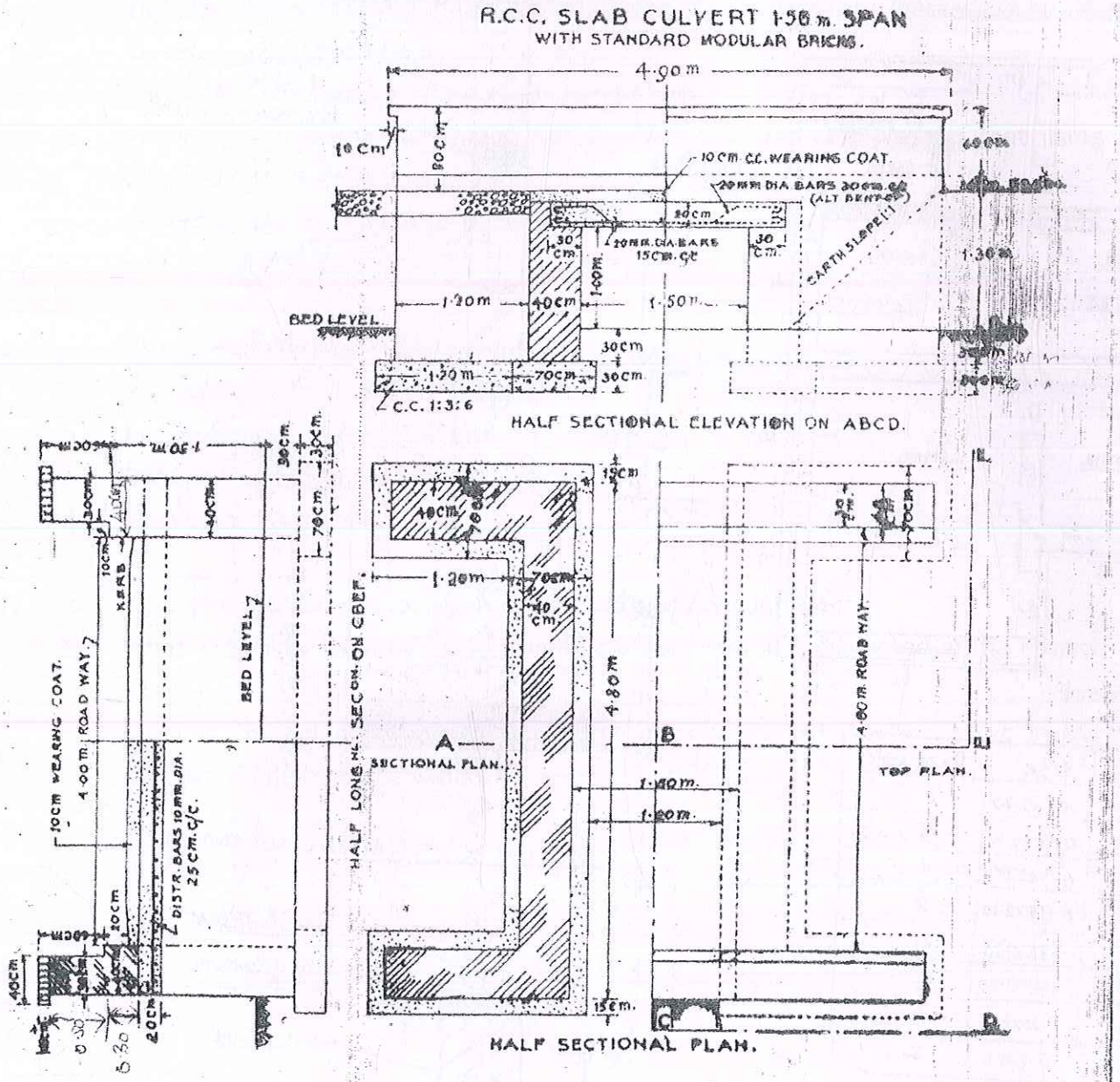
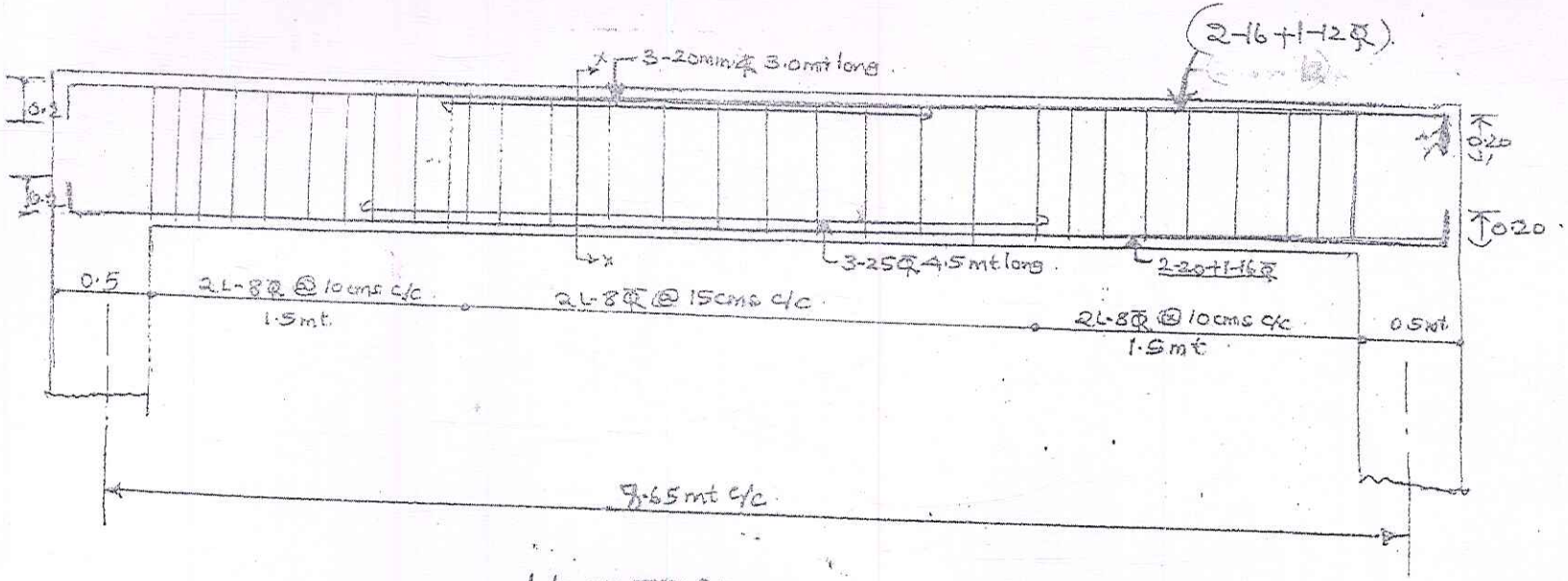
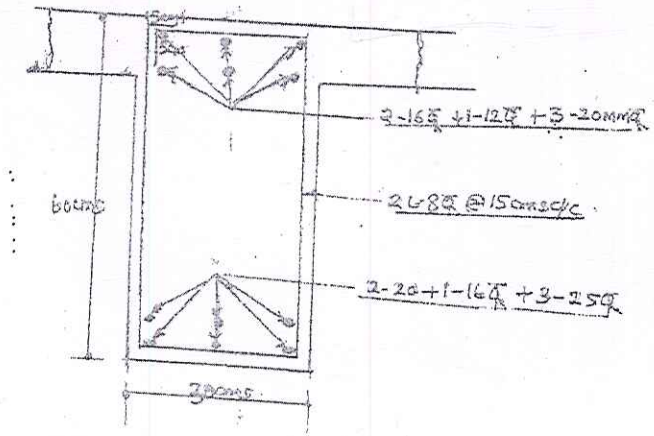

R.C.C. SLAB CULVERT 1·50 m. SPAN
WITH STANDARD MODULAR BRICKS.
4·90 m
10 Cm
80 Cm
10 cm C.C. WEARING COAT.
20 MM DIA BARS 30 cm. C/C (ALT BENT-UP)
20 cm
30 cm
20 MM. DIA BARS 15 CM. C/C
30 cm
EARTH SLOPE (1:1)
1·30 m
1·00 m
1·20 m
40 cm
1·50 m
BED LEVEL
30 cm
1·20 m
70 cm
30 cm
C.C. 1:3:6
HALF SECTIONAL ELEVATION ON ABCD.
KERB
BED LEVEL
10 CM WEARING COAT.
4·00 m ROAD WAY
DISTR. BARS 10 MM. DIA. 25 CM. C/C.
HALF LONG. SECN. ON CBEF.
A
SECTIONAL PLAN.
1·20 m
70 cm
40 cm
4·80 m
1·60 m
1·20 m
B
4·80 m ROAD WAY
TOP PLAN.
C
D
HALF SECTIONAL PLAN.

FIGURE-5



L/S OF THE BEAM 30cm x 60cm



SKETCH-3